Loucas Pillaud-Vivien

87, quai de la gare
75013 Paris
France
+33 6 77 51 33 49
loucas.pillaud-vivien@enpc.fr
https://thebiglouloup.github.io/loucaspillaudvivien/

Academic experience

2024– **Assistant Professor**, *CERMICS (UMR CNRS 9032), École des Ponts ParisTech – Institut Polytechnique de Paris*, Marne-la-Vallée, France
Member of the Hi! PARIS cluster (Institut Polytechnique de Paris – HEC Paris center in AI and data analytics).

2022–2023 **Courant Instructor – Flatiron Research Fellow**, *New York University / Flatiron Institute (Simons Foundation)*, New York, United States

2020–2022 **Postdoctoral studies in machine learning at EPFL**, *Supervisor: Nicolas Flammarion, TML*, Lausanne, Switzerland

2017–2020 **Ph.D student in machine learning in the Sierra team-project**, *Supervisors: Francis Bach, Alessandro Rudi, Inria/ENS*, Paris

2016–2017 **Mastère Spécialisé at École Nationale des Ponts et Chaussées**, *Politiques et Actions Publiques pour le Développement Durable (PAPDD)*, Marne-la-Vallée

2015–2016 **Master 2 at Université Pierre et Marie Curie (Paris VI)**, *Analyse Numérique et Équations aux Dérivées Partielles (ANEDP)*, Paris, Major in PDE (Partial Differential Equations) and Functional analysis

2015– **Ingénieur des Ponts, des Eaux et des Forêts (IPEF)**

2012–2015 **Student at École Polytechnique**, Palaiseau
Major in applied mathematics and theoretical physics

2009–2012 **Classes préparatoires**, *Lycée Louis-le-Grand*, Paris
Personal project: Random walks and Brownian motion.

2007–2009 **Lycée**, *Lycée Louis-le-Grand*, Paris
Baccalauréat in Science with Highest Honors

Teaching Experience

2025 First year research project at ENPC, (Group of 3 students).

2024 Assistant Professor of Statistics and Data Science at ENPC, (25 students during 1 semester).

2023 Research project at Courant Institute, NYU, (Group of 20 students)

2022-2023 Professor at Courant Institute, NYU, *Honors Linear Algebra* (30 elite students during 1 semester, and 250 students for Linear Algebra during 2 semesters).

Supervision

2025– **Leo Dana**, *Ph.D. student*, co-supervised with Francis Bach

2025– **Remi Surat**, *Ph.D. student*, co-supervised with Julien Guyon, CIFRE fellowship with BNP Paribas (Grégoire Loeper)

2026– **Victor Baillet**, *Ph.D. student*, co-supervised with Raphaël Berthier

2026 **Imen Ayadi**, *Postdoctoral researcher*

Student Experience

2016, 4 months **Master thesis in PDEs and Probability**, *CERMICS*, Marne-la-Vallée, Supervisors: Tony Lelièvre and Julien Reygner
Convergence of the empirical measure in a Fleming-Viot type particle system.

2015, 6 months **Master thesis in theoretical physics**, *Collège de France*, Paris, Supervisors: Antoine Georges and Michel Ferrero
Strongly correlated superconductivity studied with a Quantum Monte-Carlo algorithm and a Slave-Rotor technique.

2014, 3 months **4G Virtualization and web platform development**, *Orange Silicon Valley*, San Francisco, USA

2012–2013, 6 months **Teaching internship**, *Lycée Janson-de-Sailly*, Paris
Teaching in a social French program "Cordées de la Réussite" to promote diversity in higher education, especially in classes préparatoires.

Successes

2020 Inria-EPFL International lab Post-Doctoral fellowship "stochastic dynamics in high dimension" (co-chaired with T. Lelièvre and N. Flammarion).

Scientific Prize **Le Rivot Prize** 2015 from the **French Academy of Sciences** for the best research internship of the École Polytechnique.

Art Participation in the creation of an artistic experience for Anaïs Tondeur's art exhibition: *LOST IN FATHOMS, Nuuk Island.*

Research Grants

2025– **ANR GALA** (ANR-25-CE23-1585), Principal Investigator. French National Research Agency, approx. 400,000 EUR.

2025– **ANR DysLos** (ANR-25-CE40-6875-01), member of the consortium. French National Research Agency.

2024 **PEPS JCJC** grant, CNRS, 5,000 EUR.

Published Articles

[24] T. Jahn, L. Pillaud-Vivien, A. Schertzer
Minimax-Optimal Early Stopping for Continuous-Time SGD via the Discrepancy Principle. *Preprint*, 2026.

[23] R. Berthier, L. Pillaud-Vivien
Incremental Learning in Mirror Flows. *Preprint*, 2026.

[22] J.-C. Mourrat, L. Pillaud-Vivien
Oscillating solutions to the mean-field Langevin descent-ascent flow. *Preprint*, 2026.

[21] L. Pillaud-Vivien, A. Schertzer
Joint Learning in the Gaussian Single Index Model. *Preprint*, 2025.

[20] L. Dana, F. Bach, L. Pillaud-Vivien
Convergence of Shallow ReLU Networks on Weakly Interacting Data. *NeurIPS*, 2025.

[19] A. Schertzer, L. Pillaud-Vivien
Stochastic Differential Equations models for Least-Squares Stochastic Gradient Descent. *JMLR*, 2025.

[18] C. Margossian, L. Pillaud-Vivien, L.K. Saul.
Variational inference for uncertainty quantification: an analysis of trade-offs. *JMLR*, 2025.

[17] A. Damian, L. Pillaud-Vivien, J. Lee, J. Bruna
Computational-Statistical Gaps in Gaussian Single-Index Models. *COLT*, 2024.

[16] D. Cai et al.
Batch and match: black-box variational inference with a score-based divergence *ICML*, 2024.

[15] A. Bietti, J. Bruna, L. Pillaud-Vivien.
On Learning Gaussian Multi-index Models with Gradient Flow. Part I: General Properties and Two-timescale Learning. *Communications on Pure and Applied Mathematics*, 2025.

[14] JA Varre, M-L Vladarean, L Pillaud-Vivien, N Flammarion.
On the spectral bias of two-layer linear networks. *NeurIPS*, 2023.

[13] J Bruna, L Pillaud-Vivien, A Zweig.
On Single Index Models beyond Gaussian Data. *NeurIPS*, 2023.

[12] L Pillaud-Vivien, F Bach.
Kernelized diffusion maps. *COLT*, 2023.

[11] M Andriushchenko, A Varre, L Pillaud-Vivien, N Flammarion.
SGD with large step sizes learns sparse features. *ICML*, 2023.

[10] E Boursier, L Pillaud-Vivien, N Flammarion.
Gradient flow dynamics of shallow ReLU networks for square loss and orthogonal inputs. *NeurIPS*, 2022.

[9] L. Pillaud-Vivien, J. Reygner, N. Flammarion.
Label noise (stochastic) gradient descent implicitly solves the Lasso for quadratic parametrisation. *COLT*, 2022.

[8] S Pesme, L Pillaud-Vivien, N Flammarion.
Implicit bias of sgd for diagonal linear networks: a provable benefit of stochasticity. *NeurIPS*, 2021.

[7] V. Cabannes, L. Pillaud-Vivien, F. Bach, A. Rudi.
Overcoming the curse of dimensionality with Laplacian regularization in semi-supervised learning. *NeurIPS*, 2021.

[6] A. Varre, L. Pillaud-Vivien, N. Flammarion.
Last iterate convergence of SGD for Least-Squares in the Interpolation regime. *NeurIPS*, 2021.

[5] L. Pillaud-Vivien.
Learning with Reproducing Kernel Hilbert Spaces: Stochastic Gradient Descent and Laplacian Estimation. *Ph.D. thesis*, 2020.

[4] L. Pillaud-Vivien, F. Bach, T. Lelièvre, A. Rudi, G. Stoltz.
Statistical Estimation of the Poincaré constant and Application to Sampling Multimodal Distributions. *AISTATS*, 2019.

[3] T. Lelièvre, L. Pillaud-Vivien, J. Reygner.
Central Limit Theorem for stationary Fleming-Viot particle systems in finite spaces. *ALEA*, 2018.

[2] L. Pillaud-Vivien, A. Rudi, F. Bach.
Statistical Optimality of Stochastic Gradient Descent on Hard Learning Problems through Multiple Passes. *NeurIPS*, 2018.

[1] L. Pillaud-Vivien, A. Rudi, F. Bach.
Exponential convergence of testing error for stochastic gradient methods. *COLT*, 2018.

Academic Service

Area Chair — NeurIPS 2026.

Reviewer — *Journals:* Annals of Applied Probability, Foundations of Computational Mathematics, Annals of Statistics, JMLR, IEEE Transactions on Information Theory, SIAM Journal on Applied Dynamical Systems, SIAM Journal on Mathematics of Data Science, Applied and Computational Harmonic Analysis, IoP Science-Machine Learning: Science and Technology.

Conferences: NeurIPS, ICML, ICLR, ALT.

Languages

French — native speaker

English — fluent

Spanish, German — basics

Computer skills

Text — Microsoft Office, LaTeX

Coding — Python, PHP, HTML

Interests

Cultural activities — Contemporary Art and Theater

Sports — Surf, Tennis and Football in competition.